

```
int a = 30, b = 5, c = 2, d = 16, e = 3;
int result = a-- / ++b + c++ * --d - ++e;
```

++ / -- :Increment/decrement operator: increments or decrements data value by one.

```
int a=10;
```

```
int b=a++; //Post-increment b=10 a=11
```

```
int c=++a; //Pre-incr c=11 a=11
```

```
char x='a';
```

```
char z=x++; //z='a' x='b'
```

```
char w=++x; //w='b' x='b'
```

```
a=97
A=65
```

Handwritten calculation for the expression `int result = a-- / ++b + c++ * --d - ++e;`:

Initial values: $a=30, b=5, c=2, d=16, e=3$

Step-by-step evaluation:

- $a--$: a becomes 29, value is 29
- $++b$: b becomes 6, value is 6
- $c++$: c becomes 3, value is 3
- $--d$: d becomes 15, value is 15
- $++e$: e becomes 4, value is 4

Calculation:

$$29 / 6 + 3 * 15 - 4$$

Order of operations (PEMDAS):

$$29 / 6 = 4 \text{ (integer division)}$$

$$3 * 15 = 45$$

$$4 + 45 = 49$$

$$49 - 4 = 45$$

Final result: 45

```
int a = 10, b = 4, c = 3, d = 12, e = 2;
```

```
int result = ++a - b++ * --c + d-- / ++e;
```

A loop can perform its job by being implemented with `while`, `do-while`, or `for`.

while

While loop: maximum use in digit extraction or digit separation problems, where the control needed is more and where the exit condition is clear, the number of iterations is unclear.

do while

Maximum use for menu-driven only because this is the only loop in which, at the time of entry, we do not check the condition but at the time of exit we do.

for

Maximum use in all three loops supports iteration used in array, string, pattern printing.

Only to be used when one has all three pieces of information: the start, the condition, and iteration.

```

#include<iostream>
//1,2,3...,10
using namespace std;

int main()
{
    //for (int i=1,j=9;i<=9 && j>=1;i++,j--)
    //    cout<<endl<<i<<" " + "<<j<<" = "<<(i+j);
    for(int i=1;i<=3;i++);//If we write a semicolon at the end of the `for` loop, the entire
loop is called a single-line loop. An instruction after the loop would be treated as an
independent part of the code, not part of the loop.
    cout<<"\nhello";
    return 0;
}

```

```

#include<iostream>
/*X
XX
XXX
XXXX
XXXXX*/
using namespace std;

```

```

int main()
{
    for(int i=1;i<=5;i++)
    {
        for(int j=1;j<=i;j++)
        {
            cout<<"X";
        }
        cout<<endl;
    }
    return 0;
}

```

```

#include<iostream>

using namespace std;

int main()
{
    for(int space=5,i=1;space>=1 && i<=5;space--,i++)
    {
        //space print
        for(int s=1;s<=space;s++)
            cout<<" ";
        for(int j=1;j<=i;j++)
            cout<<"X ";
        cout<<"\n";
    }

    return 0;
}

```

```

#include<iostream>

using namespace std;

int main()
{
    for(char i='a';i<='e';i++)
    {
        for(char j='a';j<=i;j++)
        {
            cout<<j<<" ";
        }
        cout<<endl;
    }

    return 0;
}

```

```

#include<iostream>

using namespace std;

int main()

```

```
{
    int start,end;
    cout<<"Enter start and end\n";
    cin>>start>>end;
    if(start<end)//counter
    {
        for (int i=start;i<=end;i++)
        {
            cout<<i<<" ";
        }
    }
    else
    {
        for (int i=start;i>=end;i--)
        {
            cout<<i<<" ";
        }
    }

    return 0;
}
```

